Category Theory Applied to Functional Programming

Systems Engineering Undergraduate Project

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Introduction

- Category theory is a branch of mathematics developed in the 1940s.
- It is a convenient conceptual framework based on:
  - Categories
  - Functors
  - Natural transformations

Introduction

- Category theory occupies a central position in:
  - Computer science
    - Functional programming
- Several concepts of functional programming languages are based on concepts of category theory:
  - Functors
  - Parametrically polymorphic functions
  - Monads
  - Algebraic data types
  - ...

Introduction

Summary of the Project

We examine functional programming through category theory.

Introduction

Summary of the Project

Aim

- To study some of the applications of category theory to functional programming.
- In the context of:
  - Haskell
    - Functional programming language
  - Agda
    - Dependently typed functional programming language
    - Proof assistant

Introduction

Summary of the Project

Objectives

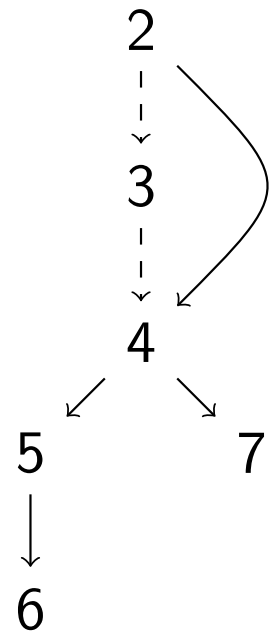
- To describe and explain the concepts of category theory needed for conceptualizing and better understanding:
  1. Functors
  2. Parametrically polymorphic functions
  3. Monads
  4. Algebraic data types

Introduction

Overview of the Project

Chapters

- Chapter 2 (Categories)
- Chapter 3 (Constructions)
- Chapter 4 (Functors)
- Chapter 5 (Natural Transformations)
- Chapter 6 (Monads and Kleisli Triples)
- Chapter 7 (Algebras and Initial Algebras)



Introduction

Overview of the Project

Chapters

- Chapter 2 (Categories)

Category theory

- Categories
- Commutative diagrams

Functional programming

- A category for Haskell
- A category for Agda

Aim and objectives

- All

Introduction

Overview of the Project

Chapters

- Chapter 3 (Constructions)

Category theory

- Isomorphisms
- Initial and terminal objects
- Products and coproducts

Functional programming

- Empty and unit types in Haskell and Agda
- Products and sums in Haskell and Agda

Aim and objectives

- All

Introduction

Overview of the Project

Chapters

- Chapter 4 (Functors)

Category theory

- Functors
- Endofunctors

Functional programming

- Functors in Haskell
- Functors in Agda

Aim and objectives

- Objective 1 (Functors)

Introduction

Overview of the Project

Chapters

- Chapter 5 (Natural Transformations)

Category theory

- Natural transformations

Functional programming

- Parametrically polymorphic functions in Haskell

Aim and objectives

- Objective 2 (Parametrically polymorphic functions)

Introduction

Overview of the Project

Chapters

- Chapter 6 (Monads and Kleisli Triples)

Category theory

- Monads
- Kleisli triples
- Monads = Kleisli triples

Functional programming

- Monads in Haskell
- Monads in Agda

Aim and objectives

- Objective 3 (Monads)

Introduction

Overview of the Project

Chapters

- Chapter 7 (Algebras and Initial Algebras)

Category theory

- Algebras and initial algebras over endofunctors

Functional programming

- Algebraic data types and folds in Haskell

Aim and objectives

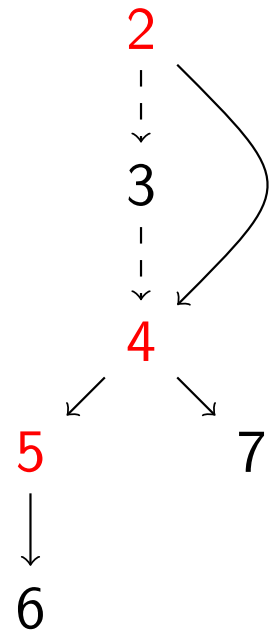
- Objective 4 (Algebraic data types)

Introduction

Overview of the Presentation

Chapters

- Chapter 2 (Categories)
- Chapter 3 (Constructions)
- Chapter 4 (Functors)
- Chapter 5 (Natural Transformations)
- Chapter 6 (Monads and Kleisli Triples)
- Chapter 7 (Algebras and Initial Algebras)



Categories

Definition (Category)

A category $\mathcal{C}$ consists of:

- Objects a, b, c, ...
- Morphisms $f : a \to b$, $g : b \to c$, ...
- Domain and codomain objects $a = \mathrm{dom}(f)$ and $b = \mathrm{cod}(f)$ for $f : a \to b$, ...
- Identity morphisms $\mathrm{id}_a : a \to a$, ...
- Composite morphisms $g \circ f : a \to c$ for $f : a \to b$ and $g : b \to c$, ...

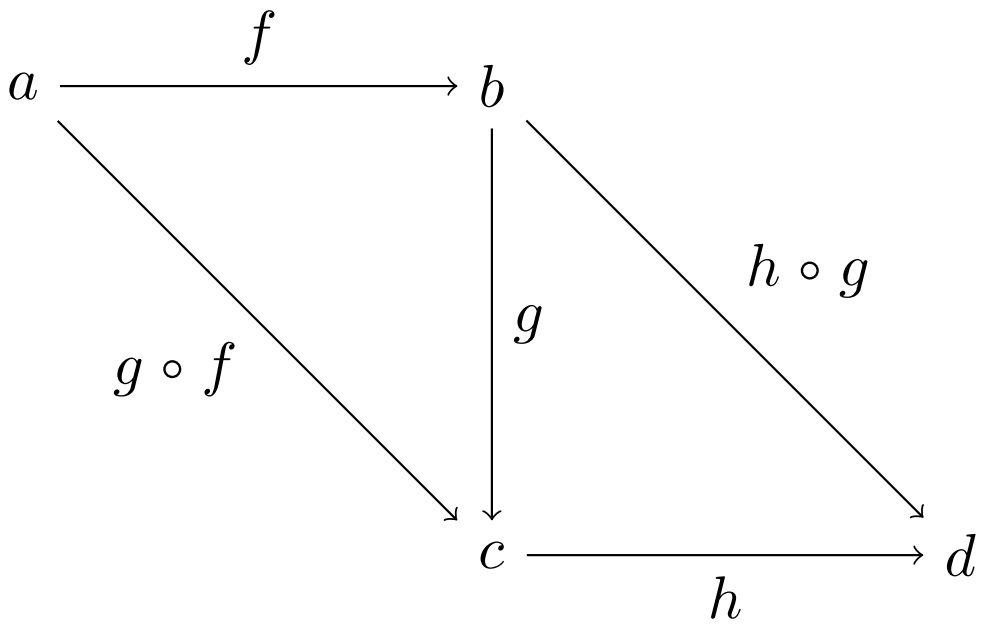
Such that:

- Associativity: $h \circ (g \circ f) = (h \circ g) \circ f$
- Identity: $\mathrm{id}_b \circ f = f = f \circ \mathrm{id}_a$

Categories

Definition (Category)

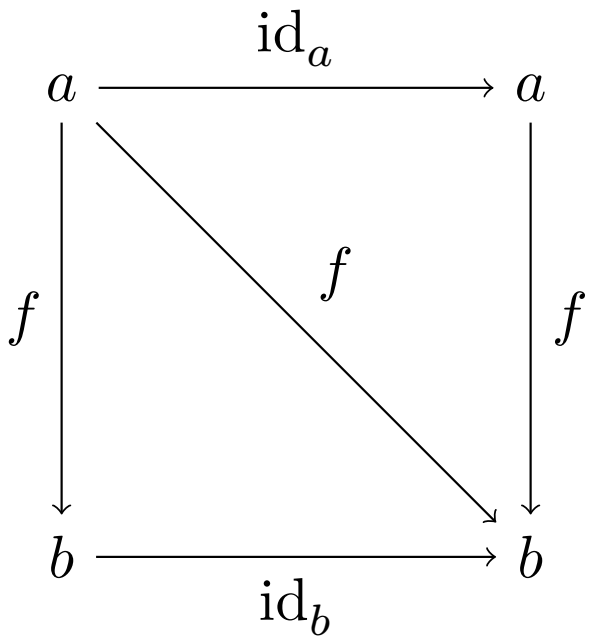
- Associativity: $h \circ (g \circ f) = (h \circ g) \circ f$



Categories

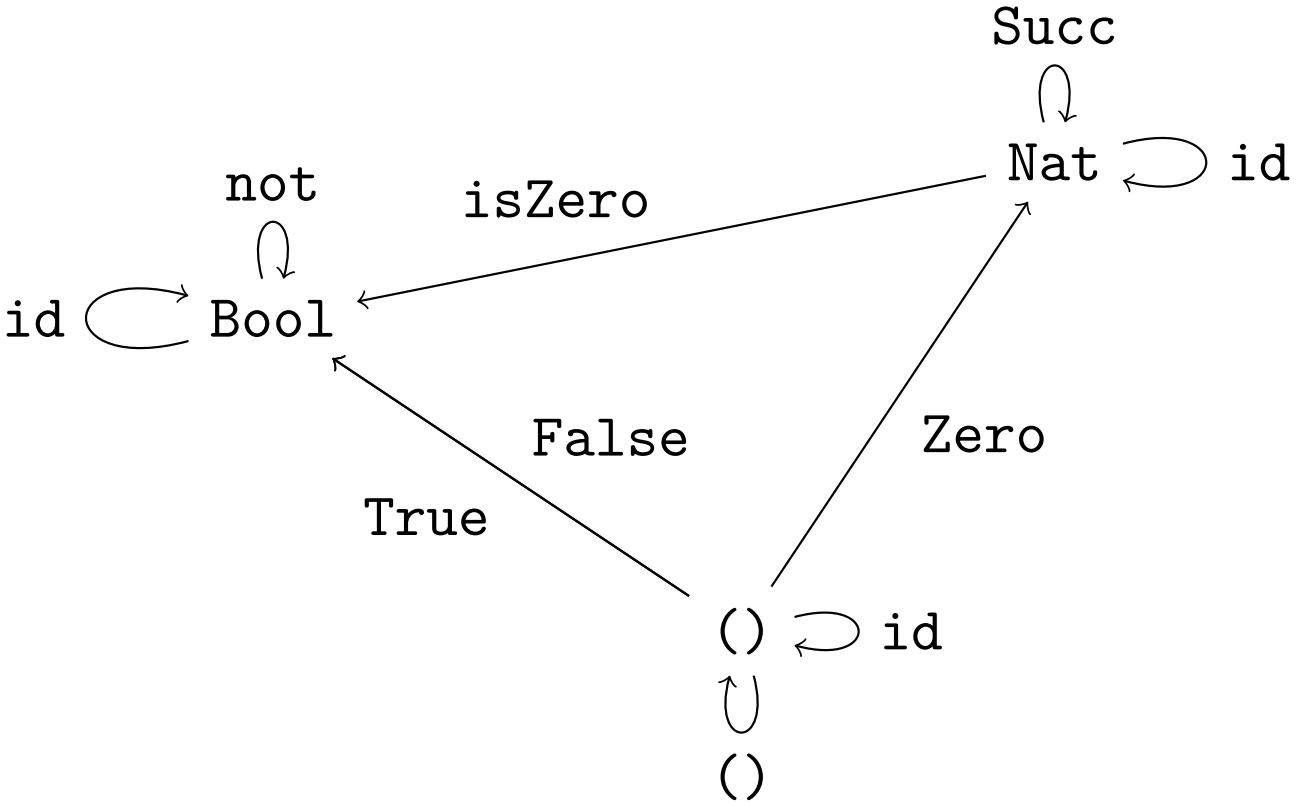
Definition (Category)

- Identity: $\mathrm{id}_b \circ f = f = f \circ \mathrm{id}_a$



Categories

A Category for Haskell



Categories

A Category for Haskell

Example

Hask is the category of Haskell types and functions.

- Objects: Haskell types with kind `*`, like `Bool`:

```
data Bool = False | True
```

- Morphisms: Haskell functions, like `not`:

```
not :: Bool -> Bool -> Bool
not False = True
not True  = False
```

Categories

A Category for Haskell

Example

Hask is the category of Haskell types and functions.

- Identities:

```
id :: a -> a
id x = x
```

- Composites:

```
(.) :: (b -> c) -> (a -> b) -> a -> c
(g . f) x = g (f x)
```

Categories

A Category for Haskell

Bottom

- Haskell types have bottom and do not yield a category.
- By Haskell types, we actually mean Haskell types without bottom, which yield a category: $\mathbf{Hask}$.

▸ Bottom

Functors

Category *has been defined in order to be able to define* functor...

—Saunders Mac Lane

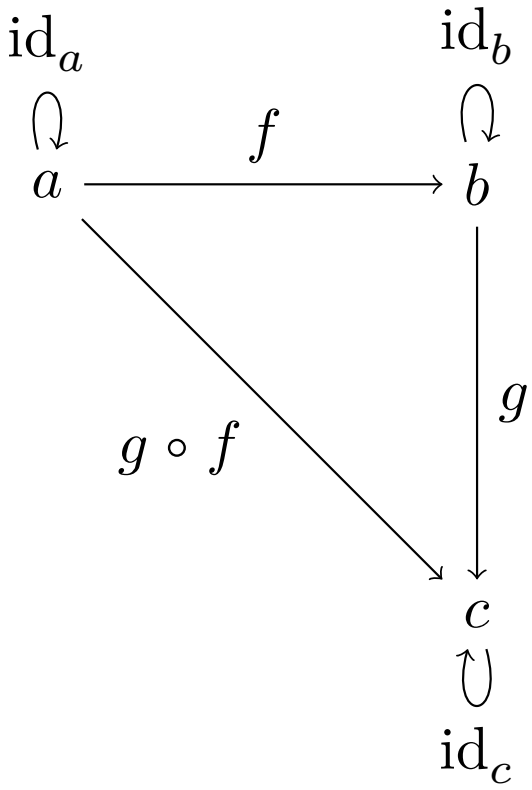
Functors

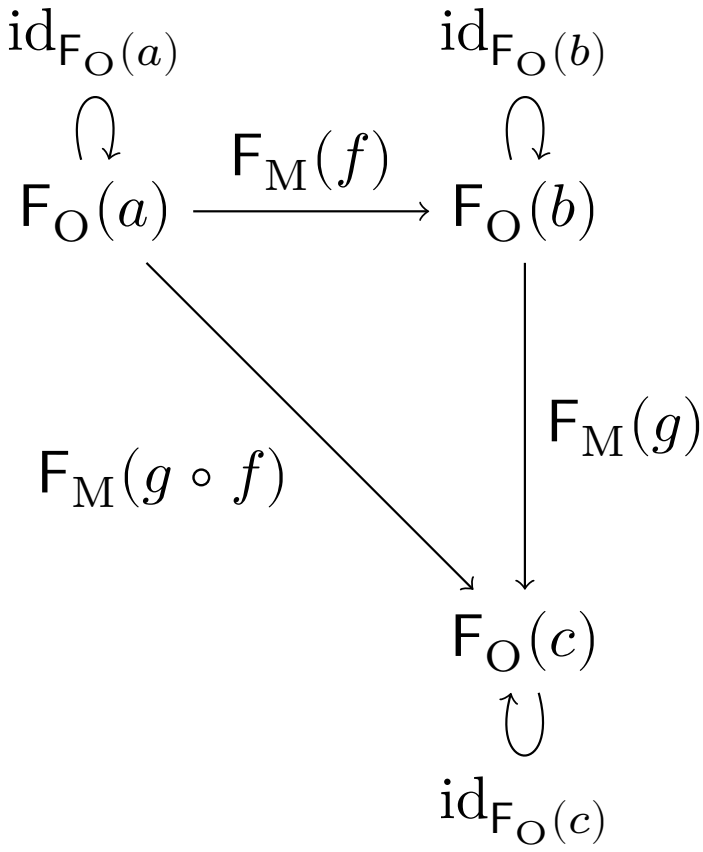
Definition (Functor)

- Let $\mathcal{C}$ and $\mathcal{D}$ be categories.
- A functor $\mathsf{F} : \mathcal{C} \to \mathcal{D}$ assigns:
  - To each object a in $\mathcal{C}$,
    an object $\mathsf{F}_{\mathrm{O}}(a)$ in $\mathcal{D}$.
  - To each morphism $f : a \to b$ in $\mathcal{C}$,
    a morphism $\mathsf{F}_{\mathrm{M}}(f) : \mathsf{F}_{\mathrm{O}}(a) \to \mathsf{F}_{\mathrm{O}}(b)$ in $\mathcal{D}$.
- Such that:
  - $\mathsf{F}_{\mathrm{M}}(\mathrm{id}_a) = \mathrm{id}_{\mathsf{F}_{\mathrm{O}}(a)}$
  - $\mathsf{F}_{\mathrm{M}}(g \circ f) = \mathsf{F}_{\mathrm{M}}(g) \circ \mathsf{F}_{\mathrm{M}}(f)$

Functors

Definition (Functor)





Functors

Definition (Endofunctor)

- Let $\mathcal{C}$ be a category.
- A functor $F : \mathcal{C} \to \mathcal{C}$ is called an endofunctor.

Functors

Functors in Haskell

Definition (`Functor`)

```
class Functor f where
  fmap :: (a -> b) -> f a -> f b
```

Functors

Functors in Haskell

Definition (`Functor`)

A `Functor` is an endofunctor in **Hask**:

```
class Functor (f :: * -> *) where
  fmap :: (a -> b) -> (f a -> f b)
```

Such that:

```
fmap id      = id
fmap (g . f) = fmap g . fmap f
```

Functors

Functors in Haskell

Example

```
data Maybe a = Nothing | Just a

instance Functor Maybe where
  fmap :: (a -> b) -> Maybe a -> Maybe b
  fmap _ Nothing  = Nothing
  fmap f (Just x) = Just (f x)
```

Functors

Functors in Haskell

Example

```
data [] a = [] | a : [a]

instance Functor [] where
  fmap :: (a -> b) -> [a] -> [b]
  fmap _ []     = []
  fmap f (x:xs) = f x : fmap f xs
```

Natural Transformations

... and functor *has been defined in order to be able to define* natural transformation.

—Saunders Mac Lane

Natural Transformations

Definition (Natural transformation)

- Let $\mathcal{C}$ and $\mathcal{D}$ be categories.
- Let F and G : $\mathcal{C} \to \mathcal{D}$ be functors.
- A natural transformation τ : F $\to$ G : $\mathcal{C} \to \mathcal{D}$ assigns:
  - To each object a in $\mathcal{C}$,
    a morphism $\tau_a : \mathsf{F}_{\mathrm{O}}(a) \to \mathsf{G}_{\mathrm{O}}(a)$ in $\mathcal{D}$.
- Such that:
  - Naturality: $\tau_b \circ \mathsf{F}_{\mathrm{M}}(f) = \mathsf{G}_{\mathrm{M}}(f) \circ \tau_a$

Natural Transformations

Definition (Natural transformation)

- Naturality: $\tau_b \circ \mathsf{F}_{\mathrm{M}}(f) = \mathsf{G}_{\mathrm{M}}(f) \circ \tau_a$

$$\begin{array}{ccc} \mathsf{F}_{\mathrm{O}}(a) & \xrightarrow{\tau_a} & \mathsf{G}_{\mathrm{O}}(a) \\ \downarrow{\scriptstyle \mathsf{F}_{\mathrm{M}}(f)} & & \downarrow{\scriptstyle \mathsf{G}_{\mathrm{M}}(f)} \\ \mathsf{F}_{\mathrm{O}}(b) & \xrightarrow[\tau_b]{} & \mathsf{G}_{\mathrm{O}}(b) \end{array}$$

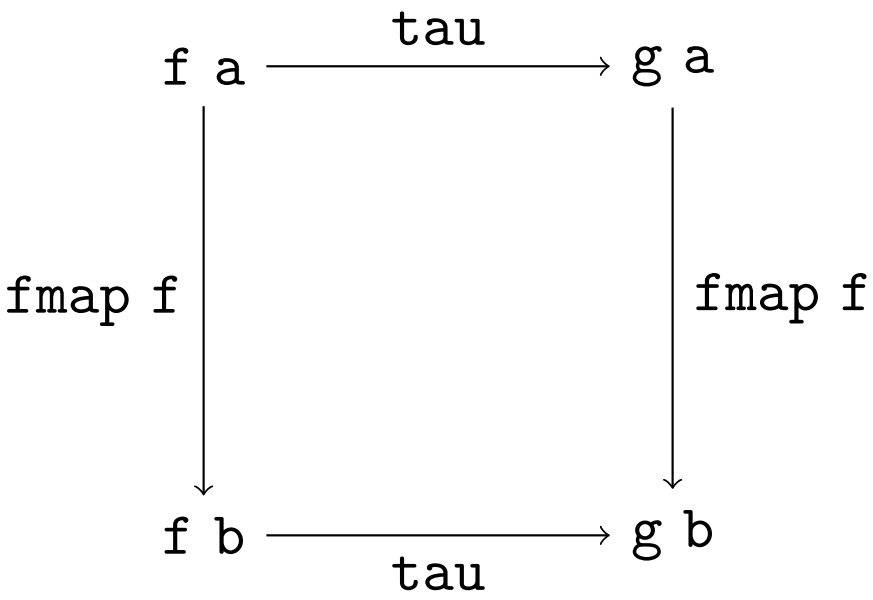
Natural Transformations

Natural Transformations in Haskell

A parametrically polymorphic function is a natural transformation.

Natural Transformations

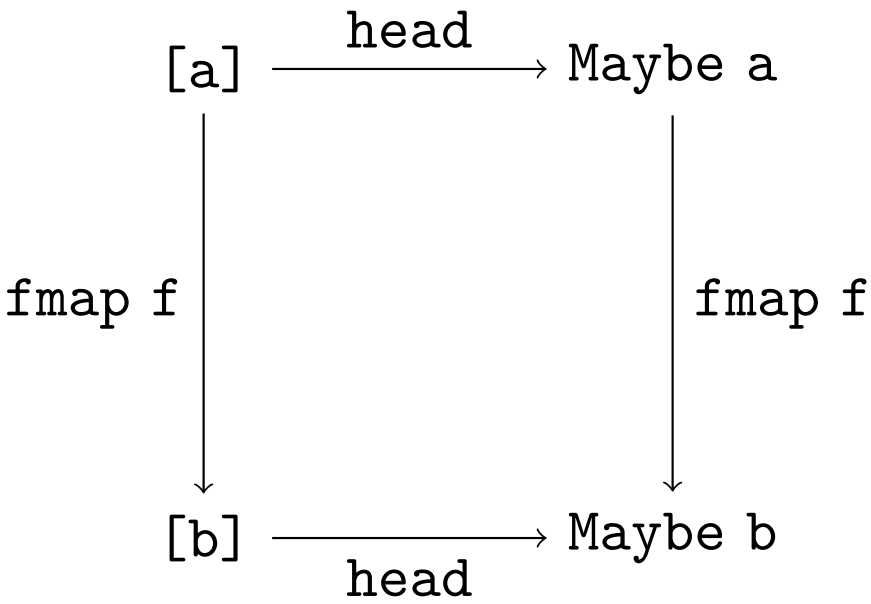
Natural Transformations in Haskell



Natural Transformations

Natural Transformations in Haskell

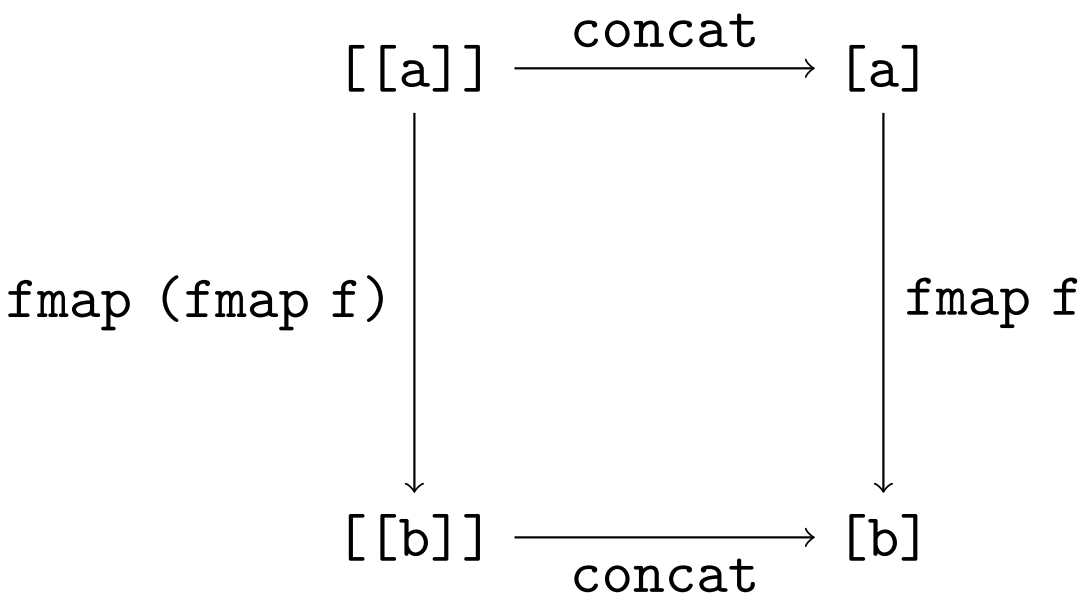
Example



Natural Transformations

Natural Transformations in Haskell

Example



Conclusions

- We did not cover all of category theory.
- But we did cover categories, functors, and natural transformations:
  - An appropriate starting point for the study of category theory applied to functional programming.
  - An overview of the project.

Conclusions

- Studying category theory may not be necessary for functional programming.
- But it is useful for:
  - Better understanding functional programming.
  - Becoming a better functional programmer.

Conclusions

- Difficult? At first.

One does not so much learn category theory as absorb it over a period of time.

—Richard Bird and Oege de Moor

- Definitely worth it.

Conclusions

Future Work

- Adjoints, which "arise everywhere."
- Applicative functors.
  - Monoidal categories and functors.
  - Haskell 2014 `Applicative => Monad` proposal.

Conclusions

Future Work

- Categories.
  - The `Category` type class.
  - The `Arrow` type class.
- Folds.
  - The `Foldable` type class.
  - The `Traversable` type class.
- Monoids, a "fundamental notion of category theory."
  - The `Monoid` type class.

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Contents

Categories

Example

Set is the category of sets and functions.

- Objects: Sets A, B, C, ...
- Morphisms: Functions $f : A \to B$, $g : B \to C$, ...
- Identities: $\mathrm{id}_A : A \to A$ such that $\mathrm{id}_A(x) = x$, ...
- Composites: $g \circ f : A \to C$ such that $(g \circ f)(x) = g(f(x))$, ...

Categories

Foundations

- To some extent, we are considering:
  - Objects of $\mathbf{Set}$: The set of all sets.
- This would lead us to a paradox such as Russell's paradox.
- We ought to assume that there is a big enough set, the universe, and take:
  - Objects of $\mathbf{Set}$: The sets which are members of the universe.

Categories

A Category for Haskell

Bottom

- Haskell types have bottom:

```
undefined :: a
undefined = undefined
```

- Haskell types do not yield a category.
- By Haskell types, we actually mean Haskell types without bottom, which yield a category: **Hask**.

See `http://www.haskell.org/haskellwiki/Hask`.

Return

Functors

Functors in Haskell

Example?

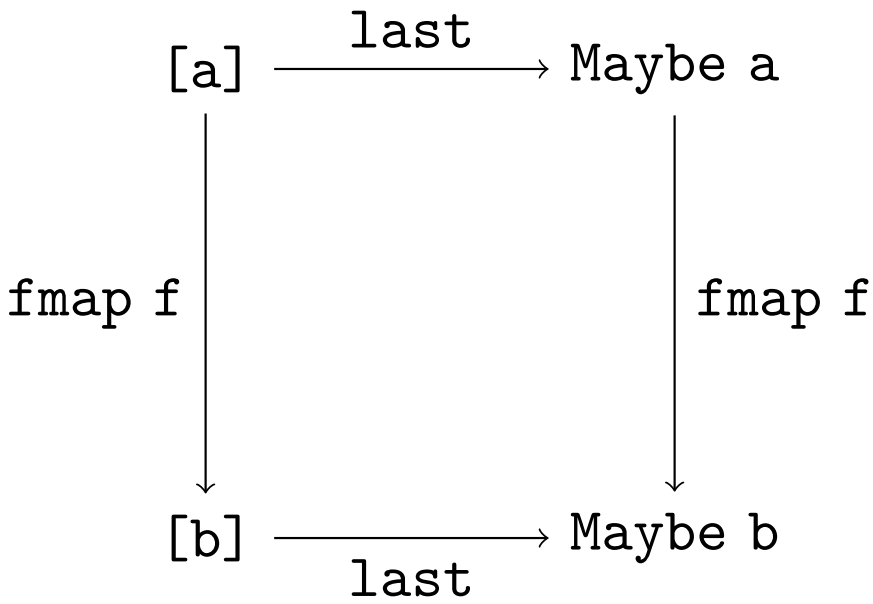
```
data [] a = [] | a : [a]

instance Functor [] where
  fmap :: (a -> b) -> [a] -> [b]
  fmap _ []     = []
  fmap f (x:xs) = f x : f x : fmap f xs
```

Natural Transformations

Natural Transformations in Haskell

Example



Monads

Definition (Monad)

- Let $\mathcal{C}$ be a category.
- A monad $\mathsf{T} = (\mathsf{T}, \eta, \mu)$ in $\mathcal{C}$ consists of:
  - An endofunctor $\mathsf{T} : \mathcal{C} \to \mathcal{C}$.
  - Natural transformations:
    - Unit: $\eta : \mathsf{I} \to \mathsf{T} : \mathcal{C} \to \mathcal{C}$
    - Multiplication: $\mu : \mathsf{T} \circ \mathsf{T} \to \mathsf{T} : \mathcal{C} \to \mathcal{C}$
- Such that:
  - $\mu_a \circ \mu_{\mathsf{T}_{\mathrm{O}}(a)} = \mu_a \circ \mathsf{T}_{\mathrm{M}}(\mu_a)$
  - $\mu_a \circ \eta_{\mathsf{T}_{\mathrm{O}}(a)} = \mathrm{id}_{\mathsf{T}_{\mathrm{O}}(a)} = \mu_a \circ \mathsf{T}_{\mathrm{M}}(\eta_a)$

Monads

Monads in Haskell

Definition (Monad')

```
class Functor m => Monad' m where
  return :: a -> m a
  join   :: m (m a) -> m a
```

Monads

Monads in Haskell

Example

```
instance Monad' Maybe where
  return :: a -> Maybe a
  return = Just

  join :: Maybe (Maybe a) -> Maybe a
  join Nothing   = Nothing
  join (Just mx) = mx
```

Kleisli Triples

Definition (Kleisli triple)

- Let $\mathcal{C}$ be a category.
- A Kleisli triple $\mathsf{T} = (\mathsf{T}_O, \eta, \_^*)$ in $\mathcal{C}$ consists of:
  - A natural transformation:
    - Unit: $\eta : \mathsf{I} \to \mathsf{T} : \mathcal{C} \to \mathcal{C}$

  And assigns:
  - To each object a, an object $\mathsf{T}_O(a)$.
  - To each morphism $f : a \to \mathsf{T}_O(b)$, a morphism $f^* : \mathsf{T}_O(a) \to \mathsf{T}_O(b)$.
- Such that:
  - $g^* \circ f^* = (g^* \circ f)^*$
  - $f^* \circ \eta_a = f$
  - $\eta_a^* = \mathrm{id}_{\mathsf{T}_O(a)}$

Kleisli Triples

Kleisli Triples in Haskell

Definition (Monad)

```
class Monad m where
  return :: a -> m a
  (>>=)  :: m a -> (a -> m b) -> m b

  ...
```

Kleisli Triples

Kleisli Triples in Haskell

Definition (Monad'')

```
class Monad'' m where
  return :: a -> m a
  bind   :: (a -> m b) -> m a -> m b
```

Kleisli Triples

Kleisli Triples in Haskell

```
Monad = Monad''
```

```
(>>=) :: Monad'' m => m a -> (a -> m b) -> m b
mx >>= f = bind f mx
```

Kleisli Triples

Kleisli Triples in Haskell

Example

```
instance Monad'' Maybe where
  return :: a -> Maybe a
  return = Just

  bind :: (a -> Maybe b) -> Maybe a -> Maybe b
  bind _ Nothing  = Nothing
  bind f (Just x) = f x
```

Monads and Kleisli Triples

Monads and Kleisli Triples in Haskell

`Monad' = Monad''`

```
bind :: Monad' m => (a -> m b) -> m a -> m b
bind f mx = join (fmap f mx)
```

Monads and Kleisli Triples

Monads and Kleisli Triples in Haskell

`Monad' = Monad''`

```
fmap :: Monad'' m => (a -> b) -> m a -> m b
fmap f mx = bind (return . f) mx

join :: Monad'' m => m (m a) -> m a
join mmx = bind id mmx
```

Algebras and Initial Algebras

- Let $\mathcal{C}$ be a category.
- Let $\mathsf{F} : \mathcal{C} \to \mathcal{C}$ be an endofunctor.

Definition (Algebra)

- An F-algebra (a, α) is:
  - An object a.
  - A morphism $\alpha : \mathsf{F}_{\mathrm{O}}(a) \to a$.

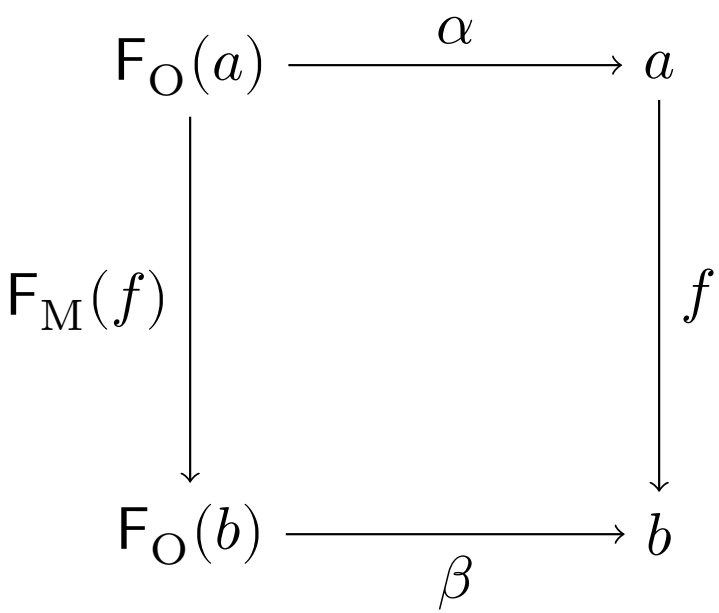
Algebras and Initial Algebras

Definition (Algebra homomorphism)

- Let (a, α) and (b, β) be F-algebras.
- An F-algebra homomorphism $f : (a, \alpha) \to (b, \beta)$ is a morphism $f : a \to b$ in $\mathcal{C}$.
- Such that:
  - $\beta \circ \mathsf{F}_{\mathrm{M}}(f) = f \circ \alpha$

Algebras and Initial Algebras

Definition (Algebra homomorphism)



Algebras and Initial Algebras

Definition

- F-$\mathbf{Alg}$ is the category with:
  - Objects: F-algebras.
  - Morphisms: F-algebra homomorphisms.

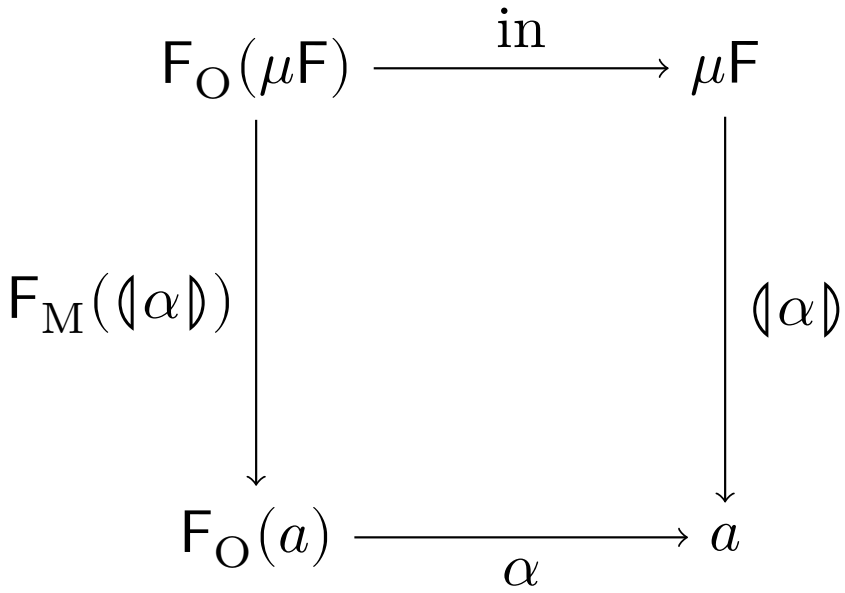
Algebras and Initial Algebras

Definition (Initial algebra)

- An F-algebra $(\mu \mathsf{F}, \mathrm{in})$ is the initial F-algebra of F-$\mathbf{Alg}$ if:
  - For all F-algebras (a, α), there is a unique F-algebra homomorphism $(\!|\alpha|\!) : (\mu \mathsf{F}, \mathrm{in}) \to (a, \alpha)$.
  - That is, a morphism $(\!|\alpha|\!) : \mu \mathsf{F} \to a$ in $\mathcal{C}$ such that:
    - $\alpha \circ \mathsf{F}_{\mathrm{M}}((\!|\alpha|\!)) = (\!|\alpha|\!) \circ \mathrm{in}$

Algebras and Initial Algebras

Definition (Initial algebra)



Algebras and Initial Algebras

Algebras and Initial Algebras in Haskell

Example

```
data [] a = [] | a : [a]
```

This is an L-algebra:

```
data L a b = N | C a b

instance Functor (L a) where
  fmap :: (b -> c) -> L a b -> L a c
  fmap _ N        = N
  fmap g (C x y) = C x (g y)
```

Algebras and Initial Algebras

Algebras and Initial Algebras in Haskell

Example

```
data [] a = [] | a : [a]
```

This is the initial L-algebra:

```
foldr :: b -> (a -> b -> b) -> [a] -> b
foldr n c []     = n
foldr n c (x:xs) = c x (foldr n c xs)
```

Algebras and Initial Algebras

Algebras and Initial Algebras in Haskell

Example

```
(++) :: [a] -> [a] -> [a]
xs ++ ys = (foldr ys (:)) xs
```

```
[]     ++ ys = ys
(x:xs) ++ ys = x : xs ++ ys
```